

During bad times, you will have much less liquidity than traditional Treasury securities. Liquidity-adjusted real rates are lower than observed real rates.

5. Hold TIPS in tax-deferred accounts.

If you want to hold linkers in an after-tax account as an individual, buy I-Bonds.

5. Commodities

Commodities are touted as natural hedges for inflation risk, especially because commodity prices enter many inflation indices.¹⁶ Commodity prices affect many items in the CPI basket. For example, oil prices directly affect gasoline, which has its own category in the CPI basket. Agricultural commodities, including wheat and corn, enter the price of food, which is another category in the CPI basket.

Commodities have become quite popular among institutional investors since the mid-2000s. Many institutions were attracted to them by the research of Erb and Harvey (2006) and Gorton and Rouwenhorst (2006), who showed that commodities had attractive returns and low correlations to stocks, bonds, and other traditional asset classes. These authors noted the good inflation-hedging ability of commodities.¹⁷ Yet, as we'll see, only certain commodities—those linked with energy—have positive correlations with inflation, and those correlations are far from perfect.

5.1. COMMODITY FUTURES

Despite the popular belief that commodities are good for hedging inflation, the baseline case in economics is that in perfect markets with no extraction costs, prices of exhaustible commodities like gold and oil should have no correlation with inflation.¹⁸ Actual prices of exhaustible commodities should follow random walks, and forward prices should rise at the risk-free rate. The intuition is that in this perfect world, producers can costlessly adjust their production to shocks and all demand shocks—whether permanent or transitory—have permanent effects on the remaining supply of these commodities. For nonexhaustible commodities, new supplies are always being created, so the normal demand and supply mechanisms, which may not involve the general price level, are at play. There is no underlying economic foundation that says commodities must be (automatically)

¹⁶ See Bodie (1983) and many others.

¹⁷ The Commodity Futures Modernization Act of 2000 also made it easier for institutional investors to take commodity positions.

¹⁸ Hotelling (1931) was the first to show this. Weinstein and Zeckhauser (1975) and Pindyck (1980) show this under uncertainty.